



Marwadi
University
Marwadi Chandarana Group

NAAC



MARWADI UNIVERSITY

FACULTY OF COMPUTER APPLICATIONS

Programming with Python & Problem Solving (05MC0107)

LAB MANUAL

Programming with Python & Problem Solving (05MC0107) Lab Manual (2026-27)

Program: MCA

Semester: 1

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Practical No. 21

Definition:

Write a program to create and import a userdefined module

Aim/Objectives:

- To understand the concept of Python modules.
- To create a module with user-defined functions.
- To save the module in a separate Python file.
- To import the module into another program.
- To call functions from the imported module.

Source Code:

[module.py](#)

```
def add(a, b):
```

```
    return a + b
```

```
def multiply(a, b):
```

```
    return a * b
```

[main.py](#)

```
import mymodule
```

```
print("Addition:", mymodule.add(10, 5))
```

```
print("Multiplication:", mymodule.multiply(10, 5))
```

```
Addition: 15
Multiplication: 50
```

Conclusion:

Thus, a user-defined module was created and successfully imported into another Python program.

Practical No. 22

Definition:

Write a program to demonstrate different import mechanisms in Python.

Aim/Objective:

- To understand the **import** statement.
- To import an entire module.
- To import specific functions from a module.
- To use aliases with the **as** keyword.
- To execute functions using different import methods.

Source Code:

```
import math

from math import sqrt

import math as m

print("Using import:", math.factorial(5))

print("Using from import:", sqrt(25))

print("Using alias:", m.pow(2, 3))
```

Output:

```
Using import: 120
Using from import: 5.0
Using alias: 8.0
```

Conclusion:

Thus, different import mechanisms in Python were demonstrated successfully.

Practical No. 23

Definition:

Write a program to perform mathematical operations using math module.

Aim/Objective:

- To understand the purpose of the `math` module.
- To calculate the square root of a number.
- To calculate the factorial of a number.
- To perform power and logarithmic operations.
- To use mathematical constants such as pi.

Source Code:

```
import math

n = 5

print("Square root:", math.sqrt(25))

print("Factorial:", math.factorial(n))

print("Power:", math.pow(2, 3))

print("Logarithm:", math.log10(100))

print("Value of pi:", math.pi)
```

Output:

```
Square root: 5.0
Factorial: 120
Power: 8.0
Logarithm: 2.0
Value of pi: 3.141592653589793
```

Conclusion:

Thus, various mathematical operations were performed successfully using the `math` module.

Practical No. 24

Definition:

Write a program to generate random numbers using random module.

Aim/Objective:

- To understand the `random` module.
- To generate random integers.
- To generate random floating-point numbers.
- To select a random element from a list.
- To shuffle the elements of a list.

Source Code:

```
import random

print("Random integer:", random.randint(1, 100))

print("Random float:", random.random())

fruits = ["Apple", "Mango", "Banana"]

print("Random fruit:", random.choice(fruits))

random.shuffle(fruits)

print("Shuffled list:", fruits)
```

Output:

```
Random integer: 42
Random float: 0.5732
Random fruit: Mango
Shuffled list: ['Banana', 'Apple', 'Mango']
```

Conclusion:

Thus, random numbers and random selections were generated successfully using the `random` module.

Practical No. 25

Definition:

Write a program to display current date and time using datetime module

Aim/Objective:

- To understand the `datetime` module.
- To obtain the current date.
- To obtain the current date and time.
- To display date and time in a readable format.
- To extract year, month, and day.

Source Code:

```
from datetime import date, datetime

today = date.today()

now = datetime.now()

print("Current date:", today)

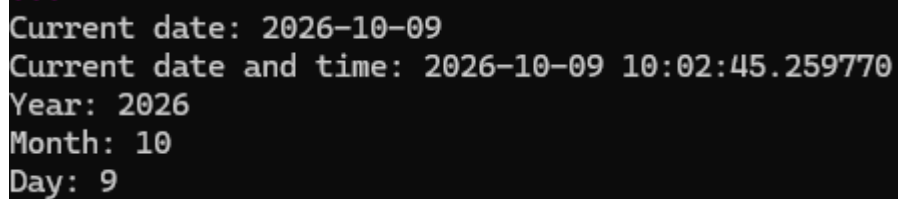
print("Current date and time:", now)

print("Year:", now.year)

print("Month:", now.month)

print("Day:", now.day)
```

Output:

A screenshot of a terminal window with a black background and white text. It shows the output of the Python program: 'Current date: 2026-10-09', 'Current date and time: 2026-10-09 10:02:45.259770', 'Year: 2026', 'Month: 10', and 'Day: 9'.

```
Current date: 2026-10-09
Current date and time: 2026-10-09 10:02:45.259770
Year: 2026
Month: 10
Day: 9
```

Conclusion:

Thus, the current date and time were displayed successfully using the `datetime` module.

Practical No. 26

Definition:

Write a program to perform file and directory operations using os and sys modules.

Aim/Objective:

To perform file and directory operations using the Python **os** and **sys** modules.

- To understand the purpose of the **os** and **sys** modules.
- To display the current working directory.
- To create a directory.
- To list files and directories.
- To display Python version and command-line arguments.

Source Code:

```
import os
import sys

print("Current directory:", os.getcwd())

if not os.path.exists("MyFolder"):
    os.mkdir("MyFolder")

print("Directory created or already exists.")
print("Files and folders:", os.listdir("."))
print("Python version:", sys.version)
print("Command-line arguments:", sys.argv)
```

Output:

```
Current directory: /home/user
Directory created or already exists.
Files and folders: ['MyFolder', 'program.py']
Python version: 3.x.x
Command-line arguments: ['program.py']
```

Conclusion:

Thus, file and directory operations were performed successfully using the **os** and **sys** modules.

Practical No. 27

Definition:

Write a program to copy move and delete files using shutil module.

Aim/Objective:

To copy, move, and delete files using the Python `shutil` module.

- To understand the purpose of the `shutil` module.
- To create a file for performing file operations.
- To copy a file to another location.
- To move a file to another directory.
- To delete a file using the appropriate function.

Source Code:

```
import os
import shutil

with open("sample.txt", "w") as f:
    f.write("Hello Python")

shutil.copy("sample.txt", "copy.txt")
print("File copied successfully.")

os.mkdir("MyFolder")
shutil.move("copy.txt", "MyFolder/copy.txt")
print("File moved successfully.")

os.remove("sample.txt")
print("Original file deleted.")

os.remove("MyFolder/copy.txt")
os.rmdir("MyFolder")
print("Copied file and folder deleted.")
```

Output:

```
File copied successfully.
File moved successfully.
Original file deleted.
Copied file and folder deleted.
```

Conclusion:

Thus, files were copied, moved, and deleted successfully using Python file-management functions.

Definition:

Write a program to demonstrate basic regular expression pattern matching.

Aim/Objective:

To demonstrate basic regular expression pattern matching in Python.

- To understand regular expressions.
- To use the `re` module.
- To identify digits in a string.
- To identify words using patterns.
- To check whether a pattern exists in a string.

Source Code:

```
import re

text = "My name is Samir and my age is 22."

if re.search(r"\d+", text):
    print("Number found")

numbers = re.findall(r"\d+", text)
print("Numbers:", numbers)

words = re.findall(r"\w+", text)
print("Words:", words)
```

Output:

```
Number found
Numbers: ['22']
Words: ['My', 'name', 'is', 'Samir', 'and', 'my', 'age', 'is', '22']
```

Conclusion:

Thus, basic regular expression patterns were used successfully to identify numbers and words in a string.

Definition:

Write a program to use re module functions such as match search and findall.

Aim/Objective:

To demonstrate the `match()`, `search()`, and `findall()` functions of the Python `re` module.

- To understand the `re` module.
- To use `match()` to check a pattern at the beginning of a string.
- To use `search()` to find the first occurrence of a pattern.
- To use `findall()` to find all occurrences of a pattern.
- To compare the results of these functions.

Source Code:

```
import re

text = "Python is easy. Python is powerful."

m = re.match(r"Python", text)
s = re.search(r"easy", text)
f = re.findall(r"Python", text)

print("Match:", m.group() if m else "Not found")
print("Search:", s.group() if s else "Not found")
print("Findall:", f)
```

Output:

```
Match: Python
Search: easy
Findall: ['Python', 'Python']
```

Conclusion:

Thus, the `match()`, `search()`, and `findall()` functions were demonstrated successfully.

Practical No. 30

Definition:

Write a program to extract specific information from a text file using regular expressions.

Aim/Objective:

To extract specific information from a text file using regular expressions in Python.

- To understand text-file handling in Python.
- To read text from a file.
- To use regular expressions to identify email addresses.
- To extract phone numbers from the file.
- To display the extracted information.

Source Code:

First store the details in a re files:

```
import re

with open("details.txt", "r") as f:
    text = f.read()

emails = re.findall(
    r"[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}",
    text
)

phones = re.findall(r"\b\d{10}\b", text)

print("Email addresses:", emails)
print("Phone numbers:", phones)
```

Output:

Email addresses: ['samir@example.com']
Phone numbers: ['9876543210']

Conclusion:

Thus, email addresses and phone numbers were extracted successfully from a text file using regular expressions.